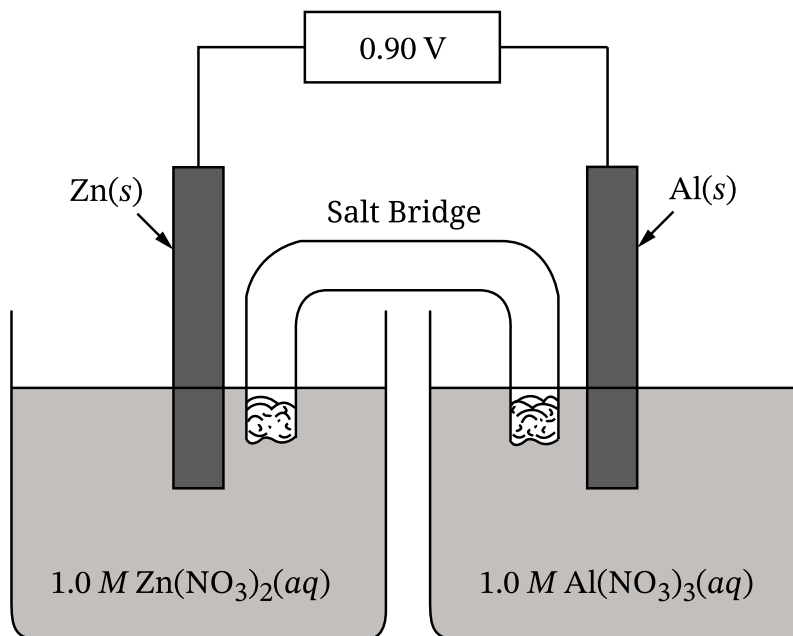


6. A scientist constructs a galvanic cell as shown in the diagram. As the cell operates, the $\text{Zn}(s)$ electrode increases in mass and the $\text{Al}(s)$ electrode decreases in mass. A data table with the standard reduction potentials for the substances follows the diagram.



Half-Reaction	E° (V)
$\text{Zn}^{2+}(aq) + 2e^- \rightarrow \text{Zn}(s)$	-0.76
$\text{Al}^{3+}(aq) + 3e^- \rightarrow \text{Al}(s)$	-1.66

- A. Write the half-reaction for the oxidation that occurs at the anode.
- B. Write the balanced net ionic equation for the overall reaction that occurs in the galvanic cell.
- C. Initially, each electrode has a mass of 50.0 g. The cell is allowed to run for a period of time and is then stopped. Which electrode's mass changed the most? Justify your answer with a calculation.